

Kundai Farai Sachikonye

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🌐 github.com/fullscreen-triangle



Bioinformatician for high-throughput omics — transcriptomics, proteomics, and metabolomics — with production AI tooling for large biological datasets, and a commitment to interdisciplinary teaching.

Education

- 2014–2017 **MSc Pharmaceutical Biotechnology**, *HAW Hamburg*
Molecular biotechnology, drug development, analytical methods.
- 2018–2019 **MSc Computational Biology**, *Universität Tübingen*
Bioinformatics of high-throughput data: sequence analysis, variant calling, RNA-seq, statistical and machine learning methods for biological datasets.
- 2019–2021 **Doctoral research, Computational Lipidomics**, *Technische Universität München*
HPC mass-spectrometry pipelines and multi-omics integration; transitioned to independent research.
- 2011–2014 **BSc Biochemistry and Cell Biology**, *Jacobs University Bremen*
Thesis: DNA interstrand crosslinking — mammalian cell culture and fluorescence microscopy.

Experience

- 2021–present **Independent Bioinformatics & Computational Biology Researcher**, *Bitspark GmbH*
Full-time development of bioinformatic pipelines and AI tooling for high-throughput omics data — genomics, transcriptomics, proteomics, and metabolomics. All systems documented, validated against reference data, and deployed as open, reproducible tools.
- 2019–2021 **Computational Lipidomics — Bioinformatics, HPC & Teaching**, *TUM / Bayerisches Forschungsinstitut*, Munich
High-throughput mass-spectrometry analysis at HPC scale; multi-omics statistical integration; processing of >100 GB datasets. Taught at Master's level and supervised student work.
- 2017–2018 **Glycomics and Proteomics**, *Universitätsklinikum Hamburg-Eppendorf*, Hamburg
Automated LC-MS/MS and MALDI-TOF data pipelines; reproducible high-throughput workflows for clinical instrument data.
- 2013 **Cancer Biomarker Discovery**, *University Medical Center Groningen*, Groningen
UPLC-MS/MS shotgun proteomics on tumour samples; analysis of high-dimensional feature sets.

Selected Bioinformatics Systems

- lavoisier **High-Throughput Mass-Spectrometry Pipeline.** Proteomics and metabolomics analysis; numerical pipeline + CNN spectral classification (PyTorch); distributed execution; 98.9% NIST accuracy on >100 GB datasets. github.com/fullscreen-triangle/lavoisier
- gospel **Multi-Omics Integration.** Joint modelling over host genotype, transcriptomics, and metagenomics; validated against 1000 Genomes, gnomAD, and UK Biobank. github.com/fullscreen-triangle/gospel

disease-state modelling **Neuroscience and Disease Characterisation.** Wearable and immuno-peptidomic disease-state models, including neurodegeneration (Parkinson's, ataxia) and tumour disease state — high-throughput signal and sequence analysis applied to pathological detection.

Technical Skills

Omics bioinformatics	Transcriptomics (RNA-seq), proteomics and metabolomics (mass spectrometry), genomics (variant calling); multi-omics integration; pathological-feature detection in high-throughput data
AI / ML for large datasets	PyTorch (CNNs, sequence models), scikit-learn, XGBoost; Bayesian and mixture modelling; foundation-model fine-tuning (LoRA); reproducible AI tooling for biological datasets
Data engineering	Ray, Dask, SLURM-compatible HPC pipelines; Docker, Kubernetes; memory-mapped I/O, HDF5, Parquet for >100 GB datasets
Languages	Python (primary), Rust, R, Bash, SQL; Nextflow / Snakemake-style workflows
Reference data	1000 Genomes, gnomAD, UK Biobank, IEDB, NIST

Teaching & Communication

Taught at Master's level at TUM (2019–2021) and supervised student work. Extensive open technical documentation and tutorial material across published bioinformatics tools; experienced in explaining high-throughput analysis to interdisciplinary, non-specialist audiences. Keen to build and deliver bioinformatics curriculum for the new Pharmaceutical & Molecular Biotechnology programme.

Languages

English	Native / Fluent (C2)
German	Conversational

resident in Germany since 2011